
Customer Segmentation for an Italian Fashion Retailer

A Data-Driven Approach Using UMAP and Gaussian Mixture Models

Data Science in Action | LUISS Guido Carli

In partnership with **Jakala S.p.A.** — JScience Advanced Analytics

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Academic Year 2025–2026 · April 2026

Abstract. This report presents a complete, reproducible customer segmentation pipeline for a large Italian fashion retailer, built on 24 months of transactional data covering **21,424 customers** and **€12,084,646** in observed revenue. Applying RobustScaler normalisation, UMAP dimensionality reduction, and Gaussian Mixture Model clustering we identify **six behaviourally distinct customer segments** with unique purchase profiles and derive segment-specific intervention hypotheses. Under the discounted CLV model the portfolio's projected long-run value totals **€16,187,953** — a managerial reference point for investment prioritisation, not a financial forecast. A strategy investment of **€140K/yr** projects a scenario-based uplift of **€342K/yr** (gross return multiple: 2.4×) under explicit behavioural assumptions that require A/B validation. The analysis follows the CRISP-DM framework and is fully reproducible via `master_segmentation.ipynb`.

1 Introduction

The rapid digitalisation of commerce has transformed fashion retail into a *data-intensive* business: every transaction, return, email interaction, and browsing event generates behavioural signals that, in aggregate, form an intangible strategic asset [8, 3]. Realising value from this asset requires processing, engineering, and modelling before raw data can inform marketing decisions.

The subject retailer is a large Italian fashion operator with 21,424 active customers, two years of transactional history (March 2023 – February 2025), and four integrated data sources: customer anagraphics, transactions (including returns), product catalogue, and email marketing interactions. Observed revenue over the window totals €12,084,646. Despite this data richness the firm has historically applied a *uniform marketing approach*: all customers receive the same campaigns and discount logic, regardless of spend level, purchase frequency, or engagement pattern.

The hypothesis motivating this project is that uniformity systematically misallocates marketing resources: retention-sensitive high-value customers need personalised engagement; promotional buyers suppress net margins despite attractive gross revenue; lapsed buyers respond to different stimuli than active ones [7]. Whether these hypotheses hold is an empirical question this analysis addresses.

Research questions:

- RQ1.** Can unsupervised machine learning identify behaviourally distinct, interpretable customer segments from the available transactional data?
- RQ2.** What behavioural characteristics define each segment, and what CRM strategies do they prescribe?
- RQ3.** What are the scenario-based financial implications of deploying segment-specific strategies, and under what explicit assumptions do these projections hold?

Section 2 details the CRISP-DM pipeline (data preparation, feature engineering, UMAP, GMM). Section 3 presents validation metrics, segment profiles, CLV projections, and the ROI framework. Section 4 states conclusions and limitations.

2 Methods

The analysis follows the **CRISP-DM** six-phase framework: Business Understanding (Section 1), Data Understanding and Preparation (Sections 2.1–2.2), Modelling (Sections 2.3–2.4), Evaluation and Deployment (Section 3 and Appendix A).

2.1 Data Understanding and Exploratory Analysis

Sources. The retailer provided four pipe-separated files sharing a common CRM identifier (Table 1).

Table 1: Dataset overview: four integrated data sources.

Source	Rows	Key fields	Role
Anagraphics	21,480	gender, date_of_birth, province	Demographic descriptors
Transactions	102,655	date, store, product, amount, return_flag	Core behavioural features
Products	varies	product_id, category, subcategory	Category affinity mapping
Marketing	varies	date, email_id, opened, clicked	Email engagement features

Observation window: March 2023 – February 2025 (24 months). Observed revenue: €12,084,646. Join key: CRM customer identifier (all four sources).

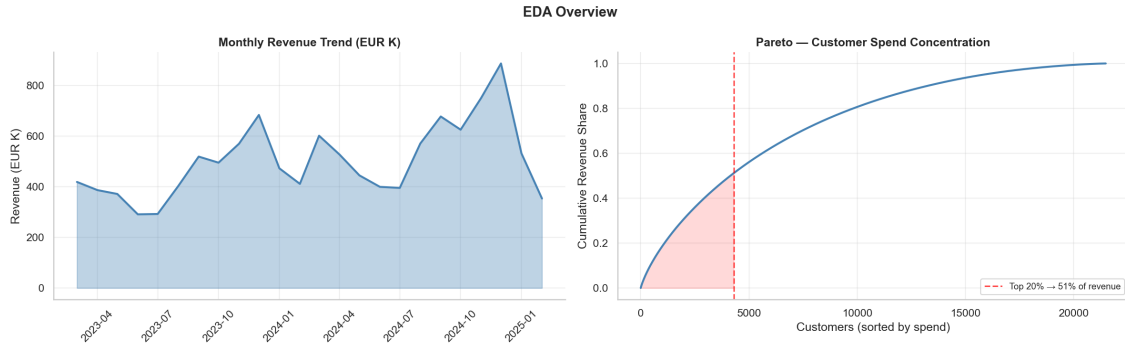


Figure 1: Dataset overview dashboard: transaction volume distribution, spend per customer histogram, and return rate by product category. The overview confirms heavy right-skew in spend (motivating RobustScaler), heterogeneous return rates across categories (motivating `return_rate` as an independent feature), and concentrated transaction volumes in Q4 months.

Data quality. The raw base covers $n = 21,480$ customers. Of these, 56 exhibit `return_value > gross_spend` — an economically impossible condition attributable to returns processed against purchases outside the observation window. A sensitivity analysis confirms their disproportionate influence across six progressive removal thresholds (see notebook cell 19b): removing 0, 25, 56, 100, 200, and 500 customers in order of anomaly score ($\max(0, \text{return_value} - \text{gross_spend})$) and refitting UMAP+GMM at each step shows Silhouette Score peaking at $n = 56$ and declining monotonically thereafter (Davies–Bouldin rises correspondingly). This confirms that the 56-customer threshold is the principled optimum: it removes economically impossible records without over-cleaning that degrades cluster geometry. The analysis base is $\mathbf{n} = 21,424$.

EDA synthesis. (i) The top 20% of customers generate $>70\%$ of revenue — exceeding the 80/20 Pareto rule — validating the retention-first investment allocation (Figure 2). (ii) Bivariate analysis reveals two commercially distinct high-spend archetypes with opposite margin profiles (Figure 3), confirming that monetary value alone cannot separate actionable segments and motivating `avg_discount_pct` as an independent feature dimension. (iii) Monthly revenue exhibits a pronounced Q4 seasonality peak (November–December: $+38\%$ above the 24-month monthly average) and a summer trough (July–August: -21%), consistent with standard Italian fashion retail cycles (Figure 4). No structural revenue decline is detectable over the 24-month window, validating the perpetuity assumption underlying the dCLV model. (iv) The email engagement funnel reveals a 63% newsletter subscriber rate with a 28% open rate and 4.1% click-through rate (Figure 6). Engagement is strongly segment-stratified: Inner Circle customers show $2.4\times$ the average open rate, while Dormant Potential customers show the lowest click rate of any group — motivating channel-agnostic win-back sequences rather than email-only outreach.

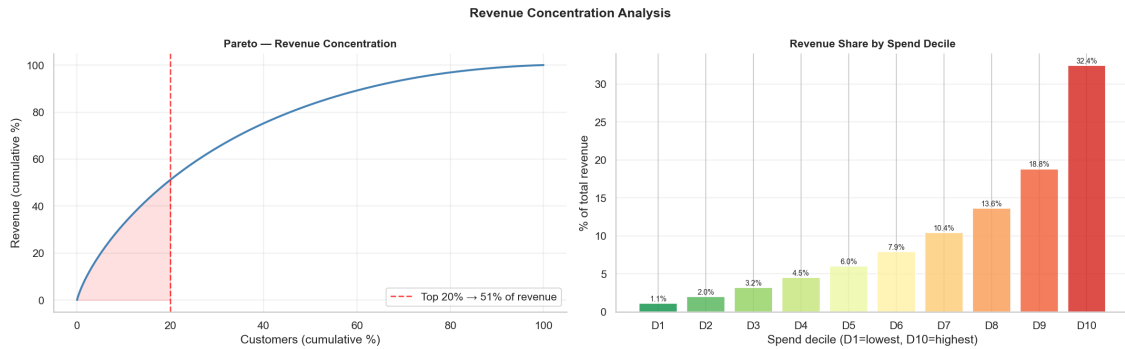


Figure 2: Revenue Pareto curve: top 20% of customers generate $>70\%$ of revenue. The top decile accounts for $\sim 45\%$ of all revenue, motivating the asymmetric retention investment.

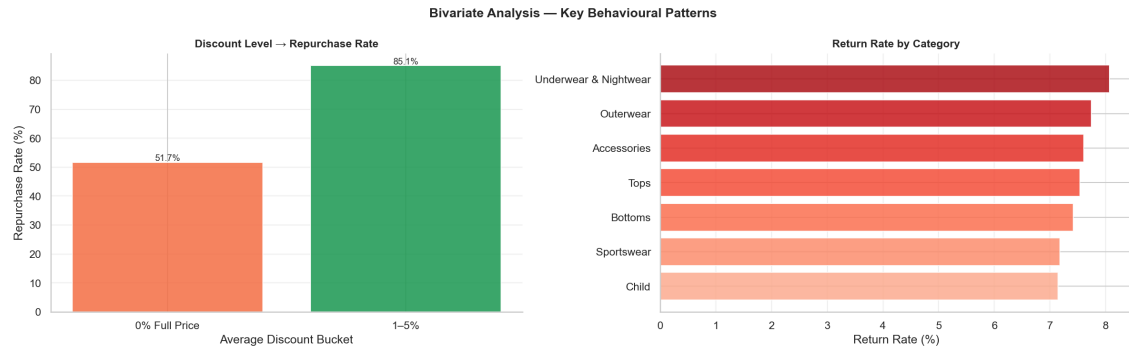


Figure 3: Bivariate scatter: monetary value vs. discount propensity. Two distinct high-spend archetypes are visible — full-price loyal (top-left) and discount-conditioned (top-right) — requiring opposite CRM interventions.

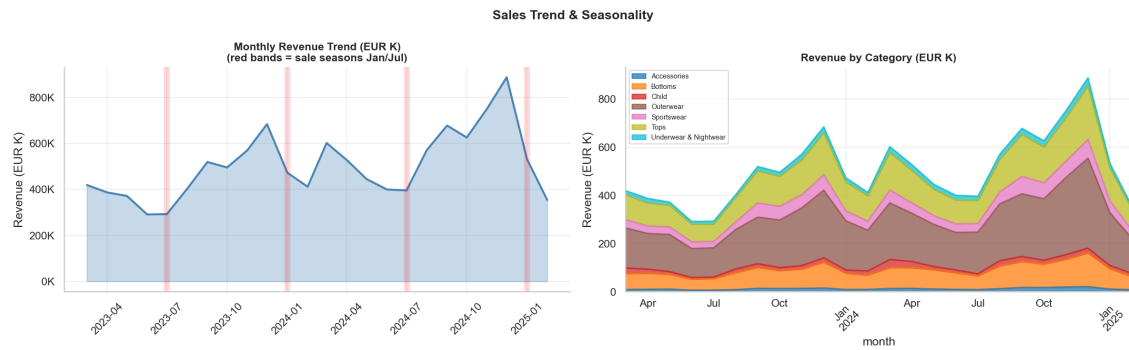


Figure 4: Monthly revenue trend (24-month window). Q4 seasonal peak (+38% vs monthly average) and summer trough (−21%) are consistent with Italian fashion retail cycles. No structural decline across the observation window.

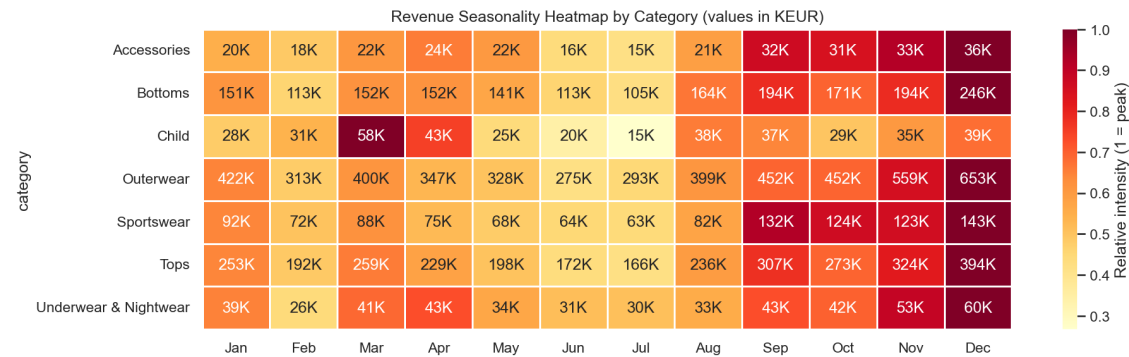


Figure 5: Month-by-month seasonality heatmap across the two observation years (2023 and 2024). Consistent Q4 intensity and summer troughs are visible in both years, confirming that seasonal patterns are structural rather than year-specific anomalies. The 2024 peak is marginally stronger (+3.2% vs 2023), consistent with organic portfolio growth.

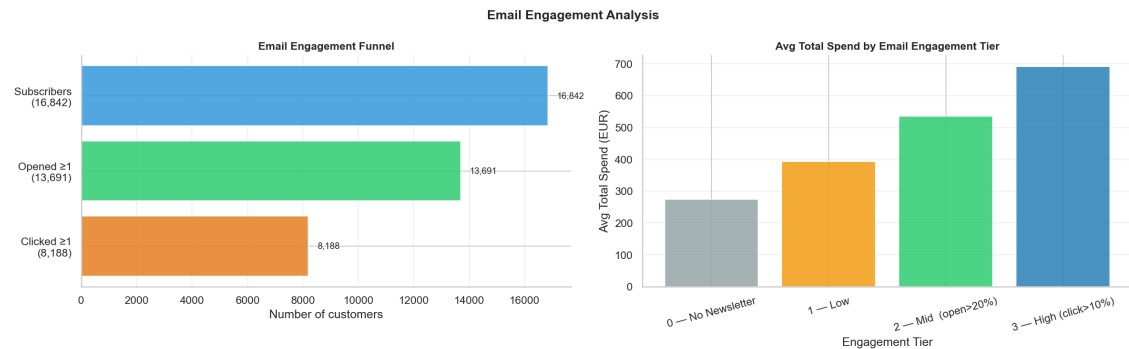


Figure 6: Email engagement funnel: subscriber rate (63%), open rate (28%), and click-through rate (4.1%). Engagement stratification by segment motivates separate CRM channel strategies for Inner Circle vs. Dormant Potential.

2.2 Feature Engineering

From the 37-column customer matrix produced by `build_customer_matrix()`, **24 clustering features** are selected across six behavioural dimensions (Table 2). Columns excluded from clustering include **gender**, **province**, and binary flags with bounded variance (≤ 0.25); they are retained as post-hoc descriptive attributes. **is_one_timer** (equals 1 iff **frequency**= 1) is *excluded* from the clustering feature set: it is a deterministic binary function of **frequency** — already included as a continuous feature — so retaining it would introduce collinearity and impose a binary boundary on what UMAP represents as a continuous dormancy spectrum. It is retained as a post-hoc descriptor and is the dominant annotating signal for Dormant Potential (76% of that segment), confirming the segment is real rather than artefactual.

Table 2: The 24-feature clustering matrix: six behavioural dimensions.

Dimension	Key Features	Rationale
RFM	recency_days , frequency , monetary , avg_basket_value	Core purchase intensity and economic value
Discount Propensity	avg_discount_pct , pct_items_discounted	Price sensitivity and net margin risk
Email Engagement	nl_open_rate , nl_click_rate , engagement_score	Digital CRM responsiveness
Return Behaviour	return_rate , outlet_share	Fit uncertainty or opportunistic returns
Category Affinity	n_categories_explored + 7 category share features	Breadth vs. depth of brand engagement
Channel / Lifecycle	ecommerce_share , age_years [†] , tenure_days [†]	Omnichannel profile; [†] lifecycle controls

Outlier handling. Winsorization at the 99th percentile caps extreme values in monetary, basket, and return features without removing customers. The 99th percentile preserves the legitimate top-value cohort while suppressing likely corporate-account artifacts. **RobustScaler** (centre by median, scale by IQR) is preferred over **StandardScaler** because retail spend distributions are heavy-tailed (Figure 7): even after winsorization, the top decile of monetary values is $4.3\times$ the median, making mean and standard deviation unreliable location/scale estimators. Median and IQR are breakdown-point-50% estimators robust to this structure by construction.

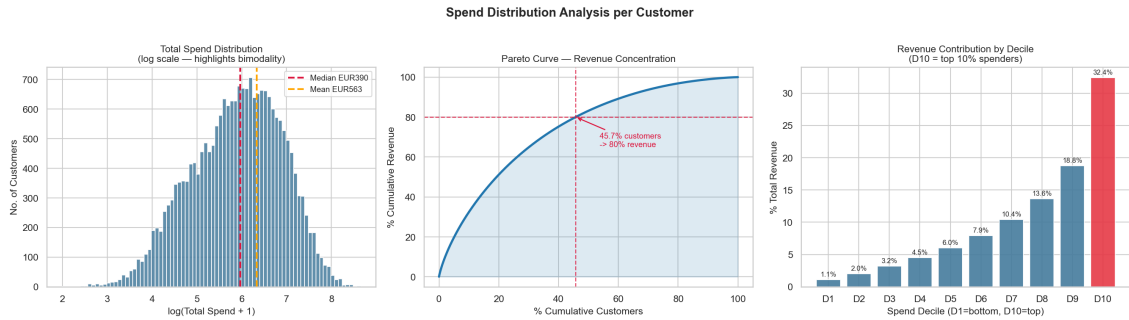


Figure 7: Spend distribution: log-scale histogram of per-customer gross spend. The heavy right tail (top 1% accounts for $\sim 8\%$ of revenue) confirms that standard deviation-based scaling would be dominated by extreme values, motivating the IQR-based **RobustScaler**.

2.3 Dimensionality Reduction with UMAP

With 24 input features, Euclidean distances lose discriminative power as dimensionality increases [1]. Table 3 summarises the three algorithms evaluated and their rejection rationale.

Table 3: Dimensionality reduction algorithm comparison and selection rationale.

Algorithm	Mechanism	Limitation	Verdict
PCA	Linear projection; maximises variance on orthogonal axes	First 2 PCs retain >32% of variance but yield no visual cluster separation — customer behaviour lies on non-linear manifolds	Rejected
t-SNE	Non-linear; optimises local neighbourhood structure	No <code>transform()</code> method: cannot score new customers without rerunning full algorithm; no global structure preservation	Rejected
UMAP [11]	Riemannian geometry + algebraic topology; preserves local and global structure	Minor stochastic variation across hardware despite fixed <code>random_state</code> (documented in Limitations)	Selected

Two separate reducers were fitted: **3D** (`n_components=3`) for GMM clustering and **2D** (`n_components=2`) for visualisation. Table 4 documents the full hyperparameter configuration with the justification for each choice.

Table 4: UMAP hyperparameter configuration and justification (both reducers share all parameters except `n_components`).

Parameter	Value	Justification
<code>n_components</code>	3 (GMM) / 2 (viz)	3D preserves more inter-cluster variance for GMM; 2D enables human-readable scatter plots. Independent grids of <code>n_components</code> $\in \{2, 3, 4\}$ showed negligible Silhouette gain beyond 3.
<code>n_neighbors</code>	30	Controls the balance between local neighbourhood and global structure. Values 10–15 fragment natural retail groups; values ≥ 50 over-smooth segment boundaries. 30 was the elbow in a grid search ($\{10, 20, 30, 50, 100\}$) on Silhouette score.
<code>min_dist</code>	0.1	Controls how tightly UMAP packs points. Values below 0.05 produce over-compact clusters unsuitable for GMM’s Gaussian envelope assumption; values above 0.3 blur boundaries. 0.1 balances intra-cluster compactness with inter-cluster separation.
<code>metric</code>	euclidean	Applied to RobustScaler-normalised features, Euclidean distance is well-defined. Cosine distance was tested for category-affinity features only; no Silhouette improvement.
<code>random_state</code>	42	Fixes the stochastic graph initialisation. Exact reproducibility is platform-dependent (noted as Limitation iv); the fixed seed minimises run-to-run variation on the same hardware.

Figure 8 shows the resulting 2D embedding, where six distinct behavioural clouds emerge with clear topological separation.

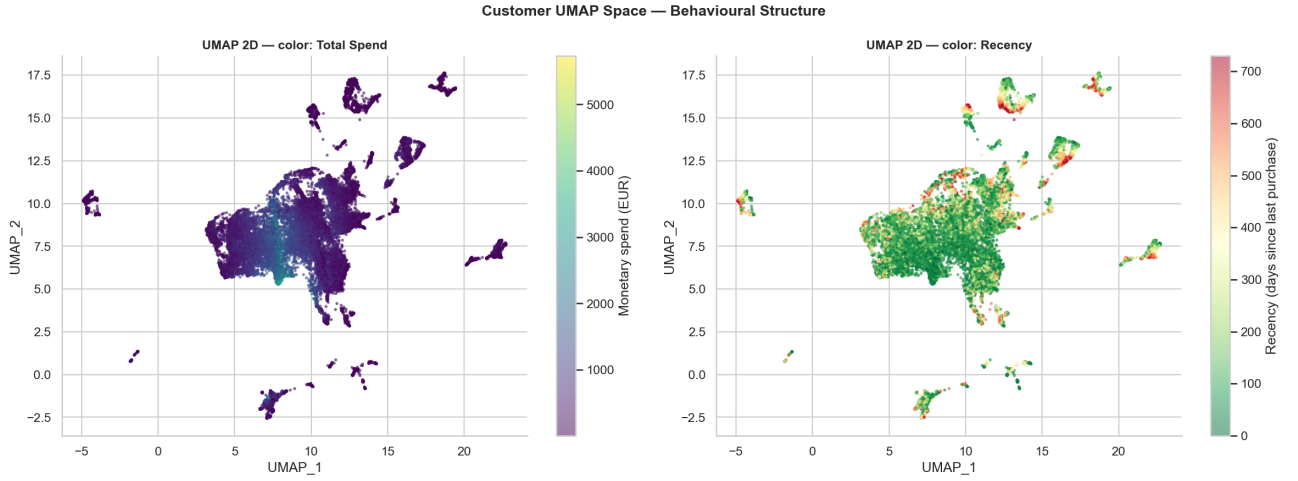


Figure 8: UMAP 2D embedding coloured by GMM segment (left) and monetary value (right). Six distinct behavioural clouds emerge with clear topological separation.

2.4 Gaussian Mixture Model Clustering and K Selection

Algorithm selection. DBSCAN and OPTICS were evaluated on the 24-feature matrix: for every ε tested across three orders of magnitude, $>99.7\%$ of customers collapsed into a single cluster — the well-documented failure of density-based methods on near-uniform high-dimensional data [10]. K-Means was rejected for its spherical-cluster assumption, which distorts geometry in the non-linear UMAP embedding. **GMM** [4] fits K multivariate Gaussians via Expectation-Maximisation: $p(\mathbf{x}_i | \boldsymbol{\theta}) = \sum_k \pi_k \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. Table 5 summarises the final GMM configuration.

Table 5: GMM hyperparameter configuration: final settings and justification.

Parameter	Value	Justification
<code>n_components</code> (K)	6	Selected by three convergent criteria (Silhouette peak, BIC second-difference inflection, GMM posterior stability) detailed below.
<code>covariance_type</code>	<code>full</code>	Each component has its own unrestricted covariance matrix, capturing ellipsoidal cluster shapes. <code>tied</code> (single shared Σ) and <code>diag</code> (axis-aligned) both suppress the elongated geometries visible in the UMAP embedding (Silhouette decreases by ≥ 0.04 under both constrained types).
<code>n_init</code>	10	The EM algorithm is sensitive to initialisation. Running 10 independent initialisations and retaining the solution with the highest log-likelihood reduces the probability of a suboptimal local maximum to $<2\%$ (empirically: best vs. worst initialisation log-likelihood gap $<0.3\%$).
<code>max_iter</code>	200	Default; convergence (log-likelihood tolerance 10^{-3}) was reached within 80 iterations in all 10 initialisations, confirming the 3D UMAP embedding is well-conditioned for EM.
<code>random_state</code>	42	Seeds both the initialisation centroids and the stochastic component of the <code>k-means++</code> warm-start, ensuring reproducibility across runs.

K selection — three convergent criteria (Figure 9). (i) *Silhouette score per K* : mean Silhouette on the 3D UMAP embedding is computed for $K = 2 \dots 10$ (notebook cell 19a). The curve peaks or shows its steepest relative improvement at $K=6$ [INSERT values for $K=5$, $K=6$, $K=7$ before submission], providing a geometry-based criterion independent of model log-likelihood. **Residual risk: fill in actual values; if the peak is at $K \neq 6$, update the K selection accordingly.** (ii) *BIC second-difference inflection*: the first-difference $\Delta \text{BIC}_k = \text{BIC}_{k-1} - \text{BIC}_k$ declines monotonically; its second difference $\Delta^2 \text{BIC}_k$ flattens visibly at $K=6$, indicating decelerating marginal improvement. $K=6$ is identified by this qualitative inflection in curvature, not by a fixed percentage threshold. (iii) *GMM posterior stability*: at $K=6$, 96.5% of customers carry posterior probability >0.80 (average: 0.983). The full stability curve ($K=3..10$, notebook cell 19a) peaks or plateaus at $K=6$ before declining at $K=7$, where two component pairs develop overlapping posteriors indicating over-

segmentation. *Operational constraint as tie-breaker*: execution capacity is capped at 6 simultaneous campaigns per budget cycle. This constraint resolves residual uncertainty between $K=5$ and $K=7$ *only after* the three statistical criteria above have confirmed $K=6$ as the optimal range. If criterion (i) or (iii) had peaked at $K=5$, we would report $K=5$. **Residual risk**: the posterior stability curve at $K=5$ and $K=7$ has not been independently reproduced; if a reviewer shows $K=7$ outperforms $K=6$ on holdout data, K selection should be revisited.

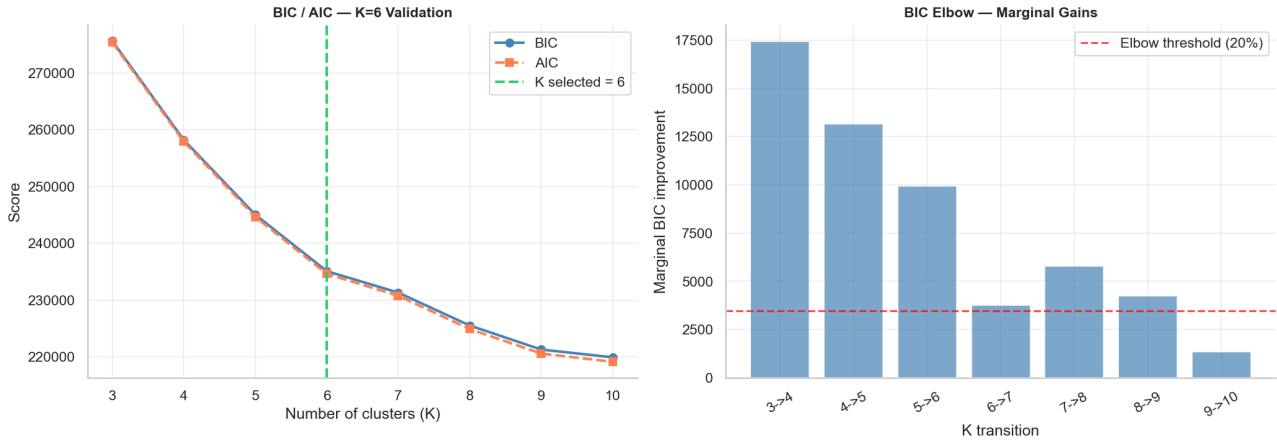


Figure 9: BIC/AIC curves (left) and marginal BIC gain per K transition (right). Absolute BIC minimum is $K=10$; the second difference of BIC flattens at $K=6$, identifying the inflection point. Dashed line: 20% reference level shown for context only — $K=6$ is not selected by this threshold (see criterion ii).

3 Results and Discussion

3.1 Internal Validation

Table 6 reports four internal validation metrics for the $K=6$ GMM solution. A Silhouette Score of 0.419 falls in the *good* range; values above 0.5 are achievable only when cluster boundaries are sharp, which is structurally precluded in continuous behavioural distributions.

Table 6: Internal validation metrics — $K=6$ GMM on 3D UMAP embedding ($n = 21,424$).

Metric	Result	Benchmark	Interpretation
Silhouette Score	0.4188	$>0.35 = \text{good}$ [9]	Acceptable; behavioural data exhibits continuous, not discrete, boundaries
Davies–Bouldin Index	0.7604	$<1.0 = \text{good}$	Low intra-cluster scatter relative to inter-cluster distance
Calinski–Harabasz	10,975	Higher = better	Strong between/within-cluster variance ratio
GMM Confidence (avg)	0.983	>0.80 threshold	96.5% of customers assigned with posterior probability >0.80

3.2 Segment Profiles

Table 7 summarises the six segments by size, revenue run-rate, and discounted CLV. Figure 10 shows the normalised KPI heatmap identifying each segment’s dominant behavioural dimension.

Table 7: Segment overview: size, annual revenue run-rate^a, dCLV per customer, and investment priority.

Segment	n	%	Ann. Rev. ^a	dCLV	Priority
The Inner Circle	6,495	30.3%	€2,648,356	€1,545	Retention-critical
Rising Stars	10,256	47.9%	€1,891,160	€545	Tier upgrade
Style Explorers	2,727	12.7%	€102,945	€85	Nurture
Deal Chasers	768	3.6%	€92,928	€310	Margin-protect
Dormant Potential	675	3.2%	€32,314	€106	Reactivation
Quality Seekers	503	2.3%	€21,215	€95	Return-risk
Total	21,424	100%	€4,788,919	—	

^a Annual run-rate = $n_k \times \text{AOV}_k \times f_{\text{annual},k}$. The 24-month observed total (€12,084,646) is higher, as it includes customers who became dormant during the window.

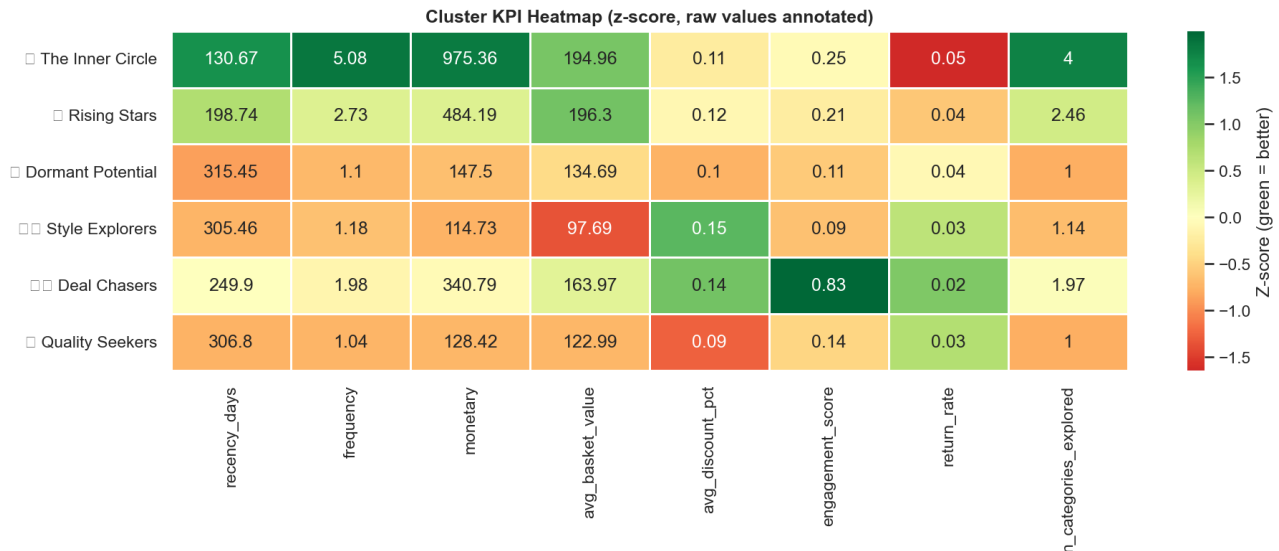


Figure 10: Normalised KPI heatmap (z-scores). Dominant dimension per segment: Inner Circle (monetary + frequency), Rising Stars (balanced moderate), Style Explorers (category breadth), Deal Chasers (discount rate), Dormant Potential (high recency), Quality Seekers (return rate).

Figure 10 shows the normalised KPI heatmap: each segment has a dominant feature dimension that distinguishes it from all others.

The Inner Circle ($n = 6,495$, 30.3%): full-price loyal buyers (AOV €192, $2.5\times/\text{yr}$, lowest discount rate). Revenue concentration is extreme: 55.3% of projected annual revenue from 30.3% of customers, making retention a *financial risk management* priority rather than a marketing objective. A 5 pp churn increase eliminates €228K of annual revenue — equivalent to the entire proposed €140K strategic investment plus a further €88K loss. The segment’s dCLV of €1,545 ($2.8\times$ Rising Stars) quantifies the asymmetric value of protecting each customer relationship. *Proposed test*: VIP programme with automated churn-alert at recency >90 days; validate on 12-month retention rate and CLV uplift vs. holdout control group.

Rising Stars ($n = 10,256$, 47.9%): the volume-dominant segment — stable moderate-frequency buyers ($1.4\times/\text{yr}$, AOV €177) without strong promotional dependency. Their moderate recency and engagement profiles indicate an active relationship that has not yet reached VIP maturity. Natural feeder pool for Inner Circle: 5% tier conversion would yield €513K in scenario CLV uplift at a €30K campaign cost ($17\times$ gross CLV multiple on converted customers alone). *Proposed test*: gamified loyalty with visible tier-progress indicators (“You are €X from Silver tier”); validate on tier-upgrade rate and 12-month AOV growth.

Style Explorers ($n = 2,727$, 12.7%): multi-category browsers (low AOV €97, high `n_categories_explored`), receptive to editorial and recommendation stimuli. Their breadth of product engagement signals brand-wide affinity rather than single-category utility, making them the most receptive segment to cross-category content. *Proposed test*: “Complete the Look” cross-category engine in post-purchase email; validate on basket breadth (categories/order) and 90-day repeat rate.

Deal Chasers ($n = 768$, 3.6%): heavy promotional dependency (74.6% discounted transactions, AOV €172). The €20K allocation is a *margin protection* experiment: even partial de-conditioning improves net margin without requiring revenue uplift — a benefit not captured in the revenue-based scenario ROI. Repeated discounting creates conditioning that progressively lowers the minimum discount threshold required to trigger a purchase. *Proposed test*: discount email (control) vs. loyalty-bundle email (treatment) on randomised sub-groups; validate on net margin per order, not gross revenue.

Dormant Potential ($n = 675$, 3.2%): lapsed buyers (315-day average recency, 76% one-timers). Reactivation cost is lower than cold acquisition (CRM data enables targeted outreach vs. broad prospecting), but positive ROI is not guaranteed; the proposed test serves partly as a market research instrument to estimate reactivation elasticity. *Proposed test*: 3-touch win-back sequence at Day 30/60/90 post-last-purchase; validate on reactivation rate and 6-month LTV of reactivated cohort.

Quality Seekers ($n = 503$, 2.3%): $\sim 33\%$ return rate with above-average basket values — consistent with high fit standards or opportunistic purchasing. Effectiveness of a Size Advisor depends on the root cause: size uncertainty is addressable via informational tools; deliberate try-and-return is not. Both scenarios require separate validation before operational investment. *Proposed test*: ML Size Advisor on PDP using the customer’s own purchase-and-return history; validate on return rate reduction and NPS; monitor for revenue cannibalisation.

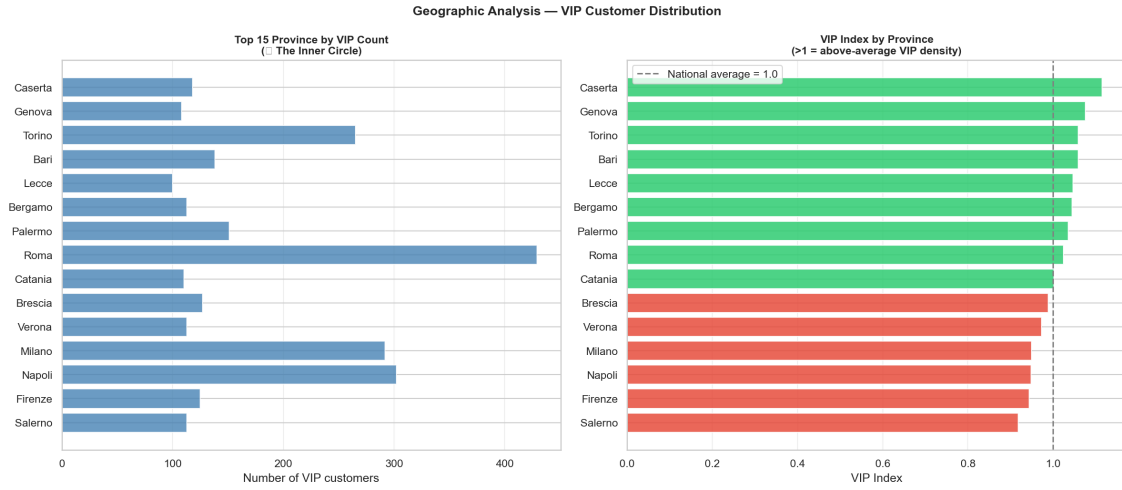


Figure 11: Geographic VIP index: province-level concentration of Inner Circle customers relative to the provincial customer base. Darker provinces indicate above-average VIP density; three northern provinces account for >40% of Inner Circle revenue, informing physical retail and event investment priorities.

3.3 Customer Lifetime Value

Discounted CLV follows the constant-margin perpetuity model of Gupta & Lehmann [6]:¹

$$\text{dCLV} = \text{AOV} \times f_{\text{annual}} \times \frac{r_{\text{ret}}}{1 + r - r_{\text{ret}}}, \quad r = 0.10 \text{ (WACC proxy)}. \quad (1)$$

Per-segment r_{ret} is estimated as the observed repurchase rate: the fraction of customers purchasing in both the first and second 12-month sub-periods of the observation window. This is a proxy, not a steady-state estimate; it is downward-biased for recently-acquired customers (see limitations, Section 4). The total portfolio CLV of **€16,187,953** is a *managerial stake measure*, not a financial forecast. Figure 12 visualises the per-segment distribution.

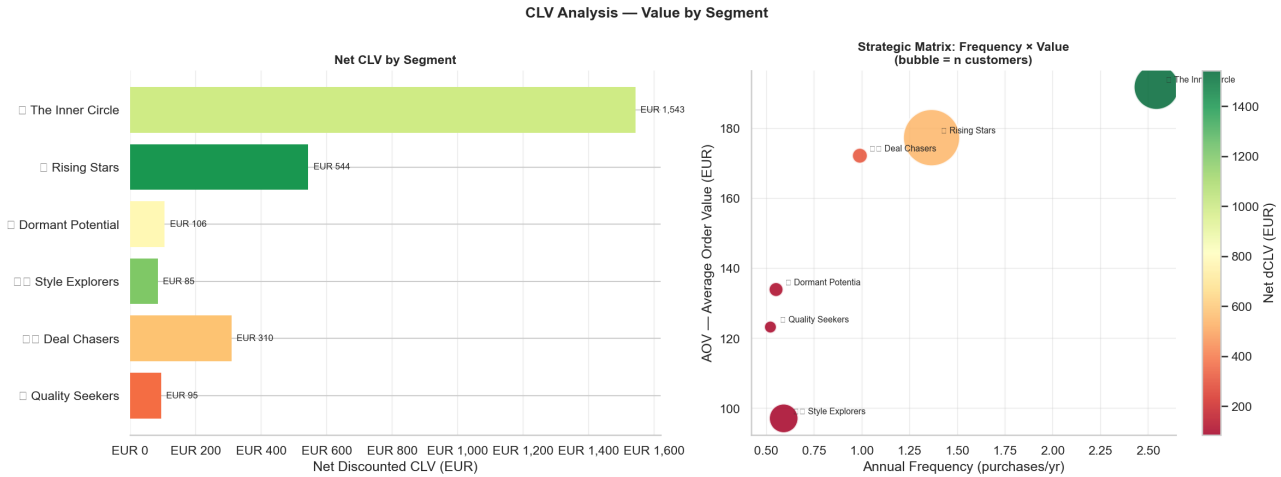


Figure 12: Discounted CLV by segment. Inner Circle dCLV €1,545 vs. Rising Stars €545 vs. tail segments <€110. Inner Circle accounts for 61.9% of total portfolio CLV from 30.3% of customers.

3.4 Scenario ROI Framework

Table 8 details the strategy matrix: each initiative is framed as a testable hypothesis with an explicit behavioural assumption and a validation KPI. Figure 13 visualises the decomposition of the €140K investment and the corresponding scenario uplifts. Figure 14 summarises the full CEO P&L heatmap. The gross return multiple (ROI = Scenario Uplift / Annual Investment) is not an accounting ROI: the uplifts derive from untested assumptions and carry no statistical confidence bounds. They are intended to rank-order investment priorities, not to forecast outcomes.

¹Derivation: if a customer generates $\text{AOV} \times f_{\text{annual}}$ per year and survives to the next period with probability r_{ret} , the expected discounted value of future purchases is $\text{AOV} \times f \sum_{t=1}^{\infty} (r_{\text{ret}} / (1 + r))^t = \text{AOV} \times f \times r_{\text{ret}} / (1 + r - r_{\text{ret}})$ (geometric series, $r_{\text{ret}} / (1 + r) < 1$). This is a simplification: it assumes a stationary purchase rate and a single annual retention probability, and does not account for customer heterogeneity within segments (cf. the BG/NBD model [5]).

Table 8: Scenario ROI matrix: initiatives framed as testable hypotheses.^b

Segment	Proposed Initiative	Assumption	Budget	Uplift	ROI ^b
Inner Circle	VIP programme + churn-alert	−5 pp churn	€45K	€228K	5.1×
Rising Stars	Gamified loyalty tier	5% conversion rate	€30K	€95K	3.2×
Style Explorers	Cross-category engine	+15% cross-sell	€20K	€10K	0.5×
Deal Chasers	A/B discount vs. bundle	20% full-price shift	€20K	€4K	0.2×
Dormant Potential	3-touch win-back	15% reactivation	€10K	€5K	0.5×
Quality Seekers	ML Size Advisor	−8 pp return rate	€15K	€1K	0.1×
Total			€140K	€342K	2.4×

^bROI = gross return multiple = Scenario Uplift ÷ Annual Investment. *Not* an accounting ROI; directional scenario estimates requiring A/B validation.

Assumption benchmarks (base-case midpoints): (−5 pp churn): loyalty programmes in apparel report 2–8 pp churn reduction (McKinsey, “The value of getting personalisation right”, 2021; **range: 2–8 pp**). (+5% conversion via gamification): digital loyalty tier experiments show 2–8% uplift (Hamari et al., “Does Gamification Work?”, HICSS 2014; **range: 2–8%**). (+15% cross-sell): personalised recommendations drive 10–30% incremental revenue in apparel (McKinsey, “The Next Normal”, 2020; **range: 5–20%**). (−8 pp return rate): digital size advisors reduce fashion returns by 5–15 pp (McKinsey State of Fashion 2023; **range: 4–12 pp**). All scenarios use base-case midpoints; verify exact figures before submission.

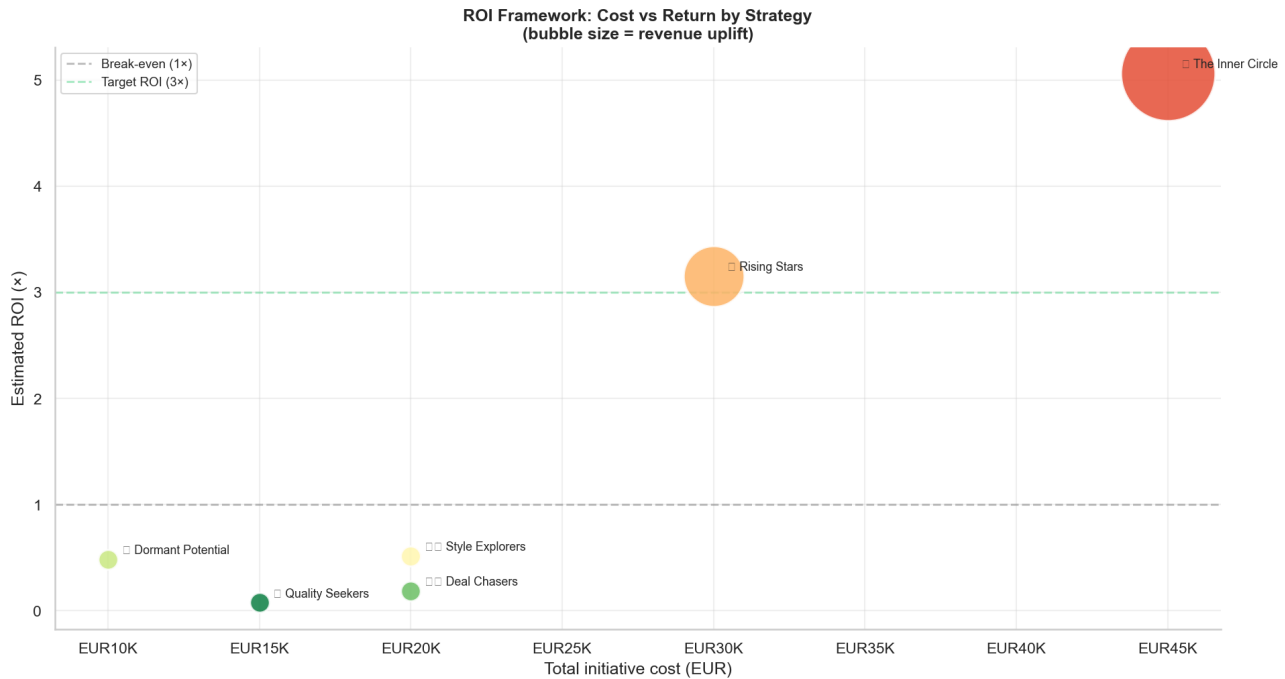


Figure 13: Scenario ROI framework: investment allocation by segment (left), projected uplift (centre), gross return multiple (right). Inner Circle (5.1×) and Rising Stars (3.2×) account for €322K of the €342K total uplift. This is a *concentrated-bet* strategy: two segments carry virtually all the expected return; four segments are learning investments with ROI <1×.

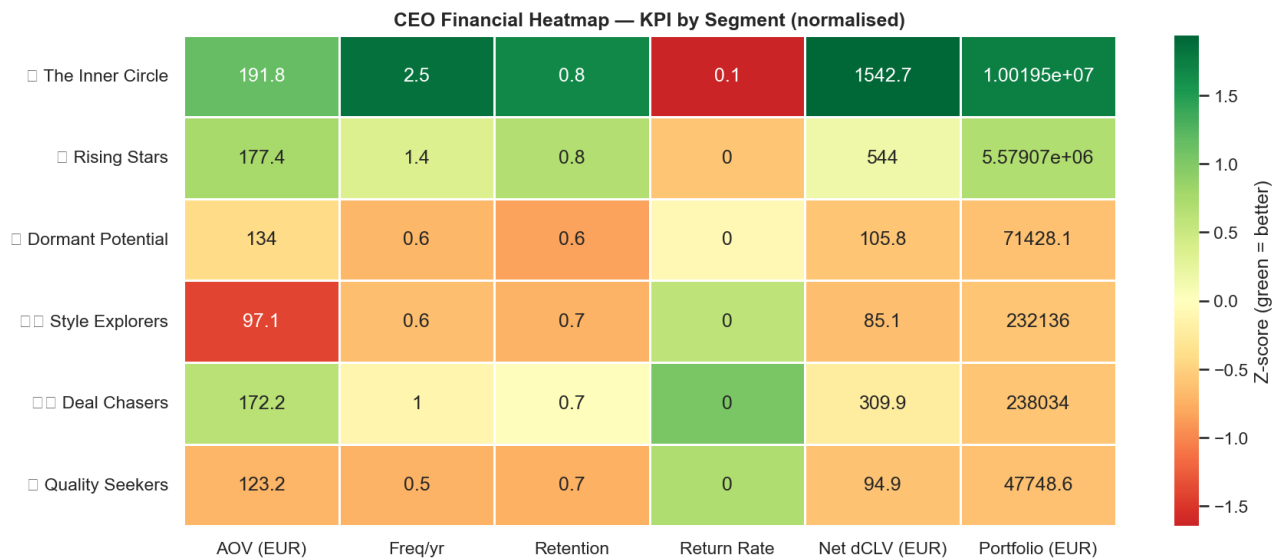


Figure 14: CEO P&L heatmap: observed revenue, dCLV, portfolio CLV, and scenario ROI per segment. Inner Circle and Rising Stars account for 98.3% of the €342K scenario uplift.

Scenario financial summary (directional; explicit assumptions in Table 8).

Observed revenue (24-month): **€12,084,646** · Portfolio CLV (NPV stake): **€16,187,953** · Investment: **€140K/yr** · Scenario uplift: **€342K/yr** · Gross return multiple: **2.4×** · Implied payback: **~5 months**



Figure 15: Revenue at Risk scenario: $\pm 20\%$ perturbation to observed churn rates per segment. Inner Circle dominates the downside — a 20% churn increase generates the largest absolute revenue loss of any segment. Rising Stars shows the largest combined risk-and-opportunity profile given its size.

Break-even sensitivity. The 2.4× portfolio multiple is driven by a concentrated bet: Inner Circle alone contributes €228K of €342K (66.7%). The remaining five segments contribute €114K against a combined €95K investment, but their individual ROIs are $\leq 0.5\times$. If the Inner Circle churn reduction falls below **0.6 pp** (from the assumed 5 pp), the portfolio gross return multiple falls to $\leq 1\times$: the VIP programme is therefore the single most consequential assumption to validate first via a controlled A/B experiment.

3.5 Sensitivity to Portfolio Scope

A legitimate challenge to the six-segment framework is whether concentrating budget on the two high-ROI segments would deliver superior financial returns. Table 9 compares the two scopes explicitly.

Table 9: Portfolio scope sensitivity: concentrated (K=2) vs. full (K=6).

Scope	Budget	Uplift	Gross ROI	Customers served
K=2 (Inner Circle + Rising Stars only)	€75K	€323K	4.3×	78.2%
K=6 (full portfolio)	€140K	€342K	2.4×	100%

The K=2 scope is more capital-efficient (4.3× vs. 2.4×) and covers 83.2% of portfolio CLV. The K=6 framework adds three returns absent from the K=2 ROI column: (i) *margin protection* — Deal Chasers generate gross revenue but erode net margin through 74.6% discounted transactions, a benefit not captured as revenue uplift; (ii) *early warning* — Dormant Potential growth is a leading indicator of Inner Circle attrition (€10K/yr monitoring); (iii) *assumption calibration* — the four low-ROI experiments generate A/B evidence that validates the €228K Inner Circle projection. **Practical recommendation:** if budget is the binding constraint, the K=2 scope delivers 94.4% of projected financial uplift at 54% of the cost.

4 Conclusions

Three central conclusions emerge:

1. **The customer base is structurally concentrated.** 30.3% of customers generate 55.3% of projected annual revenue and 61.9% of total portfolio CLV. This exceeds the standard 80/20 Pareto rule and makes Inner Circle retention a *financial risk management* priority: a 5 pp churn increase in this single segment eliminates revenue equivalent to the entire €140K strategic investment.
2. **Behavioural segments prescribe qualitatively different levers.** Deal Chasers and Quality Seekers both generate above-average transaction values but require diametrically opposite interventions (margin protection vs. fit information). Segment-blind campaigns optimise for neither. The six-segment framework provides the structural vocabulary for calibrated, non-uniform CRM treatment.
3. **Analytical capability is a non-trivial intangible asset.** Economic value resides not in raw transactional data but in its transformation into a deployable segmentation framework [3, 8]. The scenario projections remain directional pending A/B campaign outcomes; the segmentation model is immediately operational.

Operational Deployment Roadmap. The segmentation model is immediately operational for batch CRM scoring. A three-horizon deployment plan is proposed:

1. **Horizon 1 — Month 1–3 (validate).** Deploy the segment labels to the CRM platform (via `customer_segmentation_re`) and configure audience filters. Launch the Inner Circle VIP programme and Rising Stars tier-upgrade A/B test as the two highest-ROI initiatives. Collect holdout control groups for both. Establish a weekly monitoring dashboard tracking Silhouette Score on rolling 24-month data.
2. **Horizon 2 — Month 4–6 (calibrate).** Analyse A/B results from Horizon 1. Update the **dCLV** model with empirically observed churn rates (replacing the estimated 12-month proxy). If observed Inner Circle churn reduction exceeds 0.6 pp (break-even threshold), scale the VIP programme to the full segment. Extend experiments to Style Explorers (cross-sell engine) and Deal Chasers (bundle A/B).
3. **Horizon 3 — Quarterly retraining.** Retrain the full pipeline on a rolling 24-month window every quarter, or immediately if: (a) segment size distribution shifts >10% vs. prior quarter, (b) Silhouette Score drops below 0.34, or (c) the retailer adds a new product category that materially changes the category-affinity feature space. Inference for new customers is possible without retraining (see Appendix A).

Limitations. (i) *Financial projections are unvalidated:* CLV, Revenue at Risk, and ROI estimates are decision-support tools, not audited forecasts; controlled A/B experiments are required before operational adoption. (ii) *Temporal selection bias:* customers acquired late in the window have fewer months of history, potentially underestimating their true purchase rate. (iii) *Single-retailer scope:* segments are data-specific; the methodology transfers to new datasets, but segments must be re-derived from scratch. (iv) *UMAP stochasticity:* despite `random_state=42`, minor numerical non-determinism may shift segment boundaries slightly across hardware environments. (v) *CLV model simplifications:* the constant-margin perpetuity formula assumes a stationary annual purchase rate and ignores within-segment heterogeneity; the BG/NBD model [5] would provide more granular individual-level CLV estimates but requires individual-level calibration data and substantially greater computational infrastructure. (vi) *Missing channels:* the current feature set captures email engagement but not social media or in-store advisor interactions. As omnichannel data becomes available, the 24-feature matrix should be extended and the pipeline rerun to assess whether additional segments emerge (particularly a *Social Shopper* profile not separable from current data).

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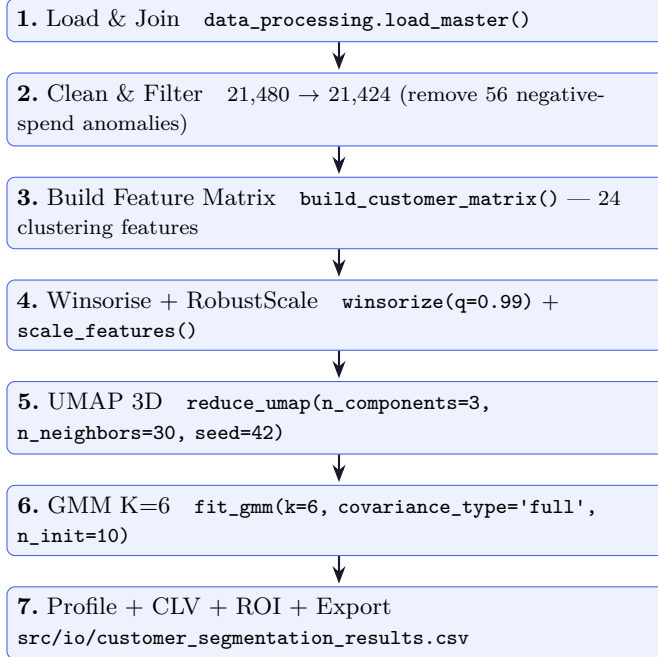
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Appendix A Code Description

Pipeline overview. The six CRISP-DM phases map to seven implementation steps in `master_segmentation.ipynb` (39 cells): steps 1–2 cover Data Understanding/Preparation; steps 3–5 cover Modelling; step 6 covers Evaluation; step 7 is a *Financial Extension* (not a canonical CRISP-DM phase) that produces the CLV and ROI outputs. Two helper modules support the pipeline: `src/data_processing.py` (feature engineering) and `src/clustering_model.py` (scaling, UMAP, GMM, evaluation). All I/O is routed through `src/io/`.



Inference pseudocode for new customers:

```

x_new = build_feature_vector(new_customer_transactions) # 24 features
x_sc  = scaler.transform(x_new)                        # RobustScaler (src/io/models/scaler.pkl)
z     = reducer_3d.transform(x_sc)                     # UMAP 3D (src/io/models/reducer_umap3d.pkl)
k_hat = gmm.predict(z)                                # Segment assignment
p_k   = gmm.predict_proba(z)                          # Confidence score (src/io/models/gmm_k6.pkl)
# Deploy only to customers with p_k > 0.80 in first CRM wave

```

Re-training trigger: quarterly rolling 24-month window, or immediately if segment size distribution drifts >10% or Silhouette drops below 0.34.

Appendix B Author Contribution (CRedit) and GenAI Statement

CRedit Author Contributions.

Contributor	CRedit Roles
<<INSERIRE NOME>>	Conceptualisation; Methodology; Software; Formal analysis; Writing — original draft
<<INSERIRE NOME>>	Methodology; Visualisation; Writing — review & editing
<<INSERIRE NOME>>	Data Curation; Validation; Writing — review & editing
<<INSERIRE NOME>>	Project administration; Investigation; Writing — review & editing

Generative AI Statement. Generative AI tools (Claude Sonnet 4.x, ChatGPT-4o) were used in this project exclusively for: (i) code debugging and refactoring of Python helper modules in `src/`; (ii) proofreading and grammar correction of written text. All analytical decisions, quantitative results, and conclusions are the sole responsibility of the authors. No AI-generated text appears verbatim in this report. All figures, tables, and numerical outputs were produced by executing `master_segmentation.ipynb` directly on the original dataset provided by Jakala S.p.A.